



Site Map · Level 2 build

Trench both ways from the panel: a second trench, drawn point by point, each one dug once. ClearSky-OMEGA

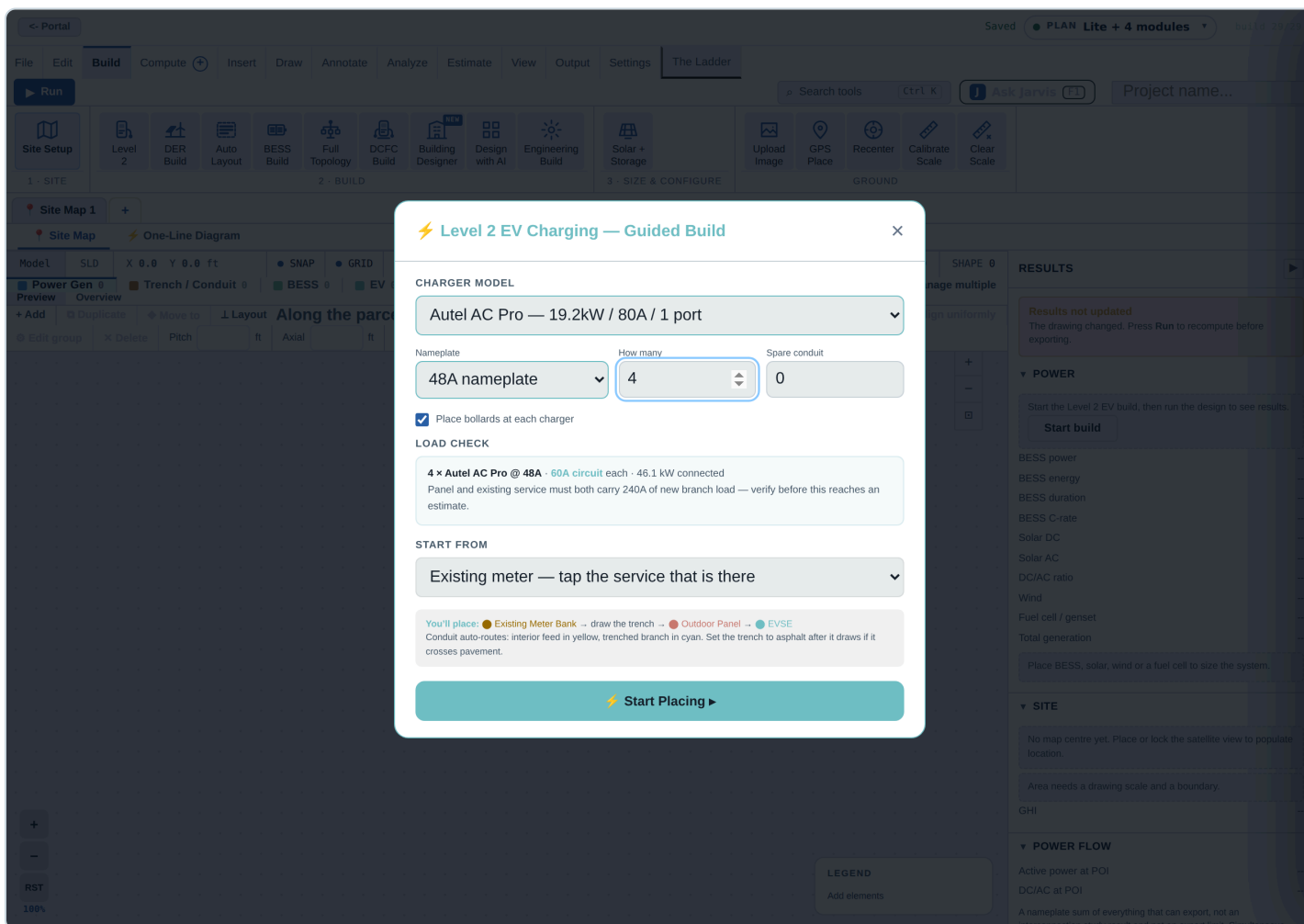
Trench both ways from the panel

A Level 2 panel on a wall often feeds stalls to its left *and* its right, or a second row behind the first. The build now draws a second trench off the first one, point by point, so the two go their own way from the panel. Each trench is dug once and nothing overlaps; every charger's conduit follows its own trench back to the panel, and the trench total counts each excavation once.

This guide runs one job from the Build tab to the finished drawing. Keys: **Enter** finishes a trench, **Backspace** removes its last point, **T** starts another trench, **Esc** backs out of that second trench.

1 Start the Level 2 build

Build tab > **Level 2**. Choose the charger model, how many, and where the service starts, then **Start Placing**. This job has four chargers.



The Level 2 dialog. Four chargers, an existing meter bank as the start.

2 Place the meter, draw the feed, place the panel

Click the map to drop the meter bank. Draw the feed run to where the outdoor panel goes (click points, **Enter**), then click on that run to place the panel. The guide panel on the right tracks each step.



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Site Map 1

One-Line Diagram

Model SLD X 110.0 Y 51.1 ft

Power Gen 0

Trench / Conduit 1

Along the parcel

Grid Behind the meter

Block out

Recalibrate

Rotate

Align uniformly

Interior feed (service-panel) [3/4" MT] . 35.1 ft . SURFACE MOUNT

Existing Meter Bank

Outdoor Panel / Disconnect

240V/120V 1Ø 3-wire — existing, verify available capacity

60A 240V 1Ø NEMA 3R · 4 × EVSE branch

LEGEND

EQUIPMENT

Existing Meter Bank

240V/120V 1Ø 3-wire — existing, verify available

Outdoor Panel / Disconnect

60A 240V 1Ø · NEMA 3R · 4 × EVSE branch

CONDUIT

Interior conduit (240V 1Ø)

35.1 ft

Draw the BRANCH trench: click points to route it, then Enter. Backspace undoes a point.

Level 2 Build

- Existing Meter Bank
- Outdoor Panel / Disconnect
- Chargers (0 / 4)

Draw the BRANCH trench: click points across the map to route it (1 pt), then press Enter to finish. Backspace removes the last point.

Finish Run (Enter)

Undo Point (Backspace)

Surface: in-ground

Cancel (Esc)

SITE

No map centre yet. Place or lock the satellite view to populate location.

Area needs a drawing scale and a boundary.

GHI

POWER FLOW

Active power at POI

DC/AC at POI

A nameplate sum of everything that can export, not an interconnection study result and not an export limit. Simultaneous

The meter, the interior feed (yellow) and the outdoor panel on its end. Next: the branch trench.



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3 Draw the first trench and drop the chargers on it

Draw the branch trench from the panel along the first row of stalls, press **Enter**, then click on the trench to drop each charger. A click past the end extends the trench to that stall.

The screenshot shows the ClearSky-OMEGA software interface for the Level 2 build. The main workspace displays a site map with a trench layout. A trench starts from the 'Outdoor Panel / Disconnect' and runs east. Two chargers, 'EVSE 1 · Autel AC Pro' and 'EVSE 2 · Autel AC Pro', are placed on the trench. A guide panel on the right shows the progress: 'Existing Meter Bank', 'Outdoor Panel / Disconnect', and 'Chargers (2 / 4)'. A legend at the bottom right lists the equipment and conduit used. A status bar at the bottom indicates 'Click on the branch trench to drop each charger (2 left) - click past its end to extend it, T draws another ...'.

The first trench heads east from the panel with two chargers on it. Two left to place.

4 Press T for another trench

Press **T**, or click **+ Draw another trench** in the guide panel. Move the mouse toward the panel: a ring shows where the new trench will begin. The first click always lands on the trench already drawn — at the panel, or anywhere along it — never beside it.



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The screenshot shows the 'Level 2 Build' interface in ClearSky-OMEGA. The top menu bar includes 'File', 'Edit', 'Build', 'Compute', 'Insert', 'Draw', 'Annotate', 'Analyze', 'Estimate', 'View', 'Output', 'Settings', and 'The Ladder'. The 'Build' tab is active, showing various tool icons for site setup, level building, DER building, auto layout, BESS building, full topology, DCFC building, building designer, design with AI, engineering build, and solar storage. The main workspace displays a site map with a trench layout. A tooltip is visible over the 'Outdoor Panel / Disconnect' equipment, showing options like 'Finish Run (Enter)', 'Undo Point (Backspace)', 'Surface: in-ground', 'Back to chargers (Esc)', and 'Cancel (Esc)'. The legend on the right lists equipment and conduit types.

LEGEND

EQUIPMENT

- Existing Meter Bank
240V/120V 1Ø 3-wire — existing, verify available
- Outdoor Panel / Disconnect
60A 240V 1Ø - NEMA 3R - 4 × EVSE branch
- Autel AC Pro x2
19 kW - 48A - 1 port
- Bollard

CONDUIT

- Interior conduit (240V 1Ø)
1 run - 35.1 ft
- Trench (panel - EVSE)
2 runs - 190.2 ft of conduit - in trench

Trench 70.0 ft - dug once

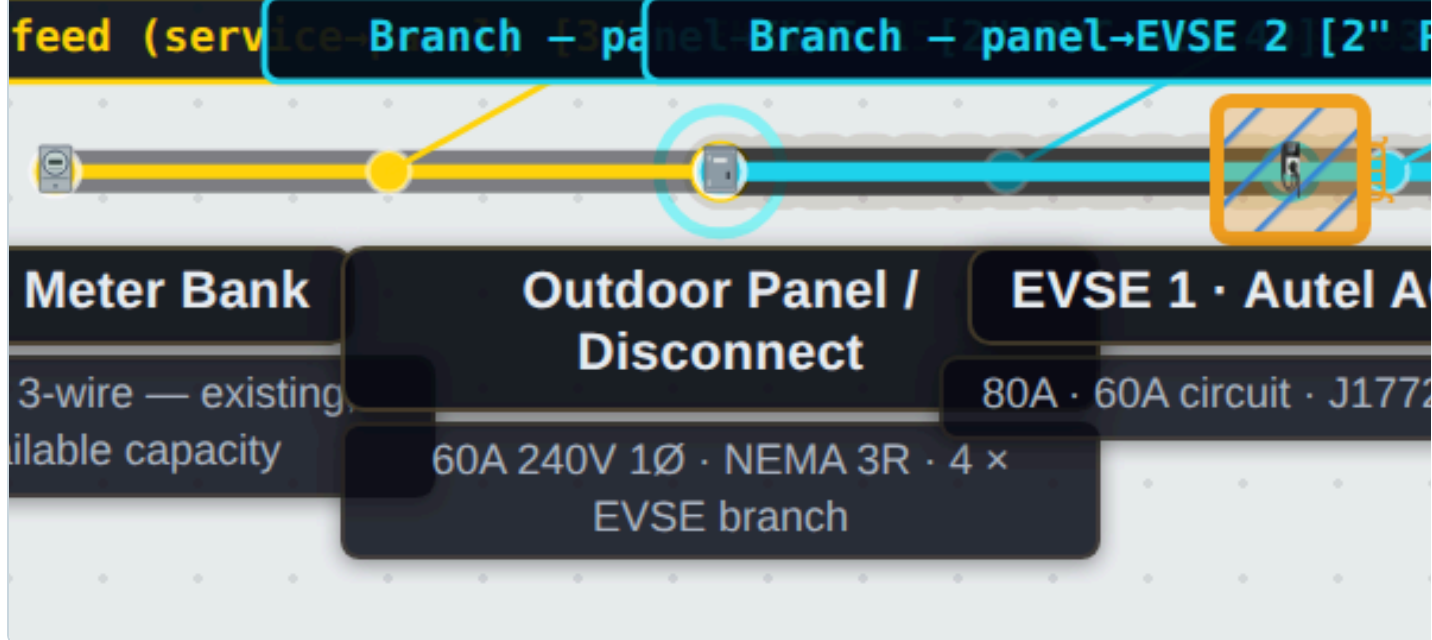
Draw another BRANCH trench: the first click lands on the trench you drew, then points to route it. Enter fin...

After T the guide panel explains the second trench; the cursor is over the panel.



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Close up: the ring on the panel is where the second trench will begin.



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5 Route it, with bends, and press Enter

Click the panel, then each bend along the way. **Backspace** takes the last point back; **Esc** or **Back to chargers** drops the whole draft. **Enter** files the trench.

Site Map 1

One-Line Diagram

Model SLD X 50.0 Y 88.6 ft

Power Gen 0

Trench / Conduit 3

BESS 0

EV 0

Wind / Gen 0

Compute 0

Fence / Tie 0

Structures 5

Manage multiple

Along the parcel

Grid Behind the meter

Block out

Recalibrate

Rotate

Align uniformly

Edit group

Delete

Pitch

ft

Axial

ft

Select an array to edit spacing

Interior feed (serv) Branch - panel-EVSE 2 [2" PVC 5ch 40] . 70.1 ft . TRENCH (IN-GROUND)

Existing Meter Bank

Outdoor Panel / Disconnect

EVSE 1 · Autel AC Pro

EVSE 2 · Autel AC Pro

240/120V 1Ø 3-wire — existing, verify available capacity

60A 240V 1Ø · NEMA 3R · 4 × EVSE branch

80A · 60A circuit · J1772 / NACS

80A · 60A circuit · J1772 / NACS

LEGEND

EQUIPMENT

Existing Meter Bank

240/120V 1Ø 3-wire — existing, verify available

Outdoor Panel / Disconnect

60A 240V 1Ø · NEMA 3R · 4 × EVSE branch

Autel AC Pro x2

19 kW · 48A · 1 port

Bollard

CONDUIT

Interior conduit (240V 1Ø)

1 run · 35.1 ft

Trench (panel - EVSE)

2 runs · 100.2 ft of conduit · in trench

Trench 70.0 ft - dug once

3 runs (the above 30 × 3.5, above and outside mounted) once · conduit

Draw another BRANCH trench: the first click lands on the trench you drew, then points to route it. Enter fin...

Level 2 Build

Existing Meter Bank

Outdoor Panel / Disconnect

Chargers (2 / 4)

Draw another BRANCH trench: your first click lands on the trench already drawn — at the panel, or anywhere along it — then click points to route it (3 pts) and press Enter. Backspace removes the last point; Esc goes back to the chargers.

Finish Run (Enter)

Undo Point (Backspace)

Surface: in-ground

Back to chargers (Esc)

Cancel (Esc)

DC/AC ratio

Wind

Fuel cell / genset

Total generation

Place BESS, solar, wind or a fuel cell to size the system.

SITE

No map centre yet. Place or lock the satellite view to populate location.

Area needs a drawing scale and a boundary.

GHI

POWER FLOW

Active power at POI

DC/AC at POI

A nameplate sum of everything that can export, not an interconnection study result and not an export limit. Simultaneous

The second trench drawn south from the panel, then west: three points, still a draft (the guide panel counts them).



The diagram illustrates the power distribution for the EVSE installation. It starts with an **Existing Meter Bank** (240/120V 1Ø 3-wire — existing, verify available capacity). The main line is a **Branch — panel** (2" PVC Sch 40) running 70.1 ft in a **TRENCH (IN-GROUND)**. This branch splits into two paths:

- EVSE 1 · Autel AC Pro**: 80A · 60A circuit · J1772 / NACS. This unit is connected to a **Outdoor Panel / Disconnect** (60A 240V 1Ø · NEMA 3R · 4 x EVSE branch).
- EVSE 2 · Autel AC Pro**: 80A · 60A circuit · J1772 / NACS.

The diagram also shows a **Branch — panel-EVSE 2** (2" PVC Sch 40) running 70.1 ft in a **TRENCH (IN-GROUND)** to the second EVSE unit.

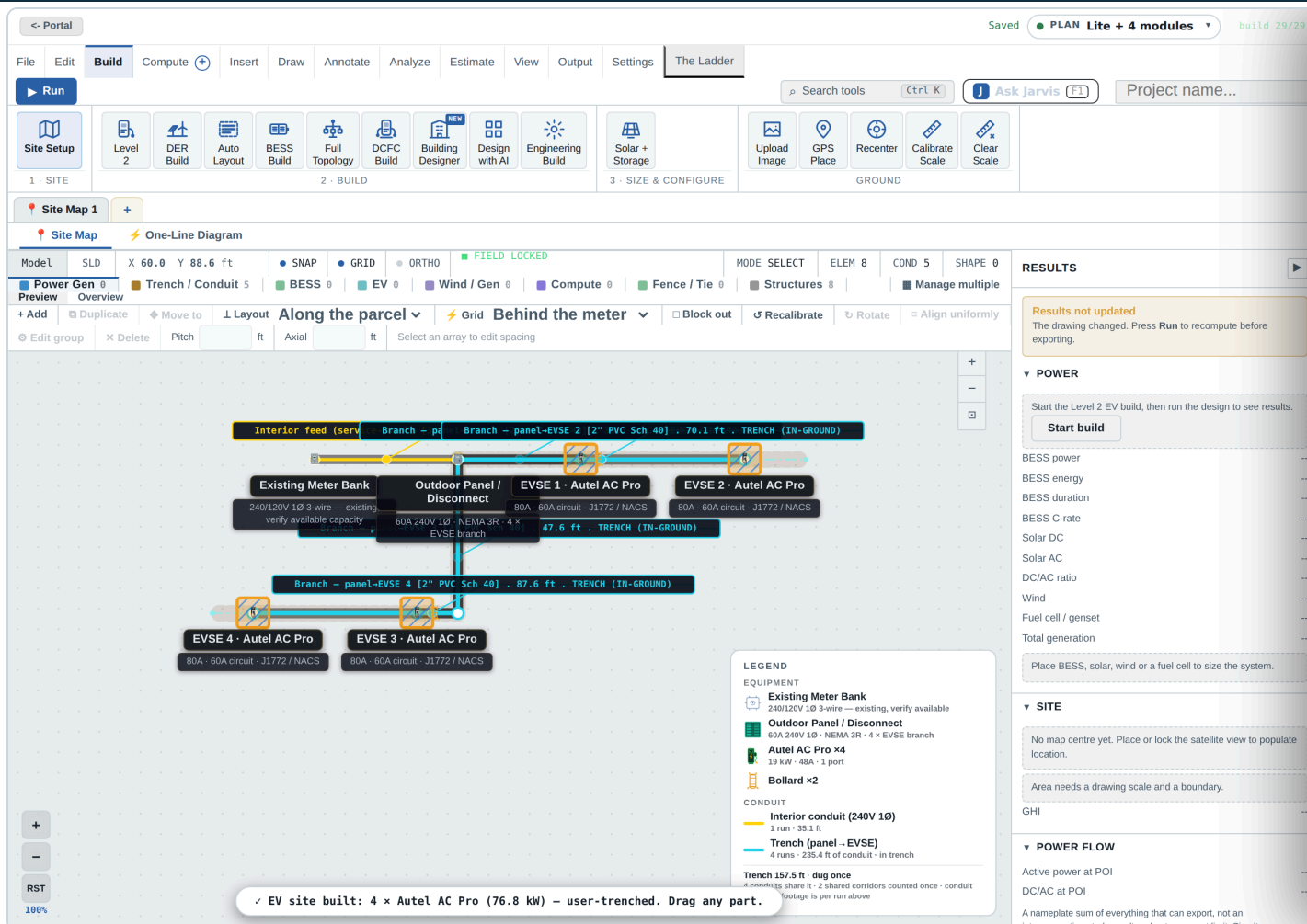
6 Drop the rest of the chargers on the new trench

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Four chargers on two trenches leaving the same panel. Nothing overlaps, nothing is dug twice.



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7 What the totals say

The legend counts the trench as excavation: each drawn trench once, the first trench only as far as the second one taps it, and the conduit inside them separately. The conduit schedule and the BOM read the same numbers.

LEGEND

EQUIPMENT



Existing Meter Bank

240/120V 1Ø 3-wire — existing, verify available



Outdoor Panel / Disconnect

60A 240V 1Ø · NEMA 3R · 4 × EVSE branch



Autel AC Pro ×4

19 kW · 48A · 1 port



Bollard ×2

CONDUIT



Interior conduit (240V 1Ø)

1 run · 35.1 ft



Trench (panel → EVSE)

4 runs · 235.4 ft of conduit · in trench

Trench 157.5 ft · dug once

4 conduits share it · 2 shared corridors counted once · conduit and cable footage is per run above

The legend after the build: the branch conduit runs, and the trench dug once.

Good to know



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- A charger clicked *off* the trench still gets a straight spur from the nearest point. Use **T** when the route has to bend before the first stall, or should leave from a particular point.
- The second trench can start anywhere on the first one, not only at the panel: a tee off the middle of a row works the same way.
- The surface button (in-ground or sawcut concrete) applies to the trench being drawn.
- **Esc** while drawing the *second* trench goes back to the chargers; while drawing the first trench it cancels the build, keeping what was placed.
- The same control is in the DCFC build.